

by construction

$Sv = 0$ and every ceiling hold exactly at v^*; the pool holds exactly on E_g

feasible for every genotype by construction: the zero flux with full slack satisfies every row and bound

a genotype that cannot meet its maintenance pays M per unit unmet, so z^* goes negative

arm flux_projection_anchored_nodiss: no dissipation hinge; Gibbs dissipation is logged only

measured (round 18 calibration)

calibration through the layer at initialization: at ceiling scale 1 the maintenance shortfall is 0.54 of 0.70 mmol/gDW/h with zero growth; from scale 10 no slack; a gene-owned bound carries a dual on 97 percent of genotypes at scales 30 to 3,000